Introduction to Quantum Information and Quantum Machine Learning

Lab 4

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Introduction

This lab shows the implementation of different forms of Grover's oracle and the Grover diffusion operator. The XOR and phase oracles are presented along with Grover's Diffusion operators and the associated mathematical derivations

Imports

In [1]:
```
import math
from math import sqrt
from numpy import pi
from qiskit import *
from qiskit.quantum_info import Statevector, Operator
from qiskit import QuantumRegister, ClassicalRegister, QuantumCircuit
from qiskit_aer import AerSimulator
from qiskit.visualization import *
import numpy as np
```

Task 1 :

In [2]:
```
backend = AerSimulator(method='unitary')
experiment_backend = AerSimulator()
```

XOR Oracle

In [3]:
```
xor_oracle = Operator([[1, 0, 0, 0, 0, 0, 0, 0],
                       [0, 0, 0, 0, 0, 1, 0, 0],
                       [0, 0, 1, 0, 0, 0, 0, 0],
                       [0, 0, 0, 1, 0, 0, 0, 0],
                       [0, 0, 0, 0, 1, 0, 0, 0],
                       [0, 1, 0, 0, 0, 0, 0, 0],
                       [0, 0, 0, 0, 0, 0, 1, 0],
                       [0, 0, 0, 0, 0, 0, 0, 1],])
Operator.is_unitary(xor_oracle)
```

Out[3]: True

Phase Oracle

In [4]:
```
phase_oracle = Operator([[1, 0, 0, 0, 0, 0, 0, 0],
                       [0, 1, 0, 0, 0, 0, 0, 0],
                       [0, 0, 1, 0, 0, 0, 0, 0],
                       [0, 0, 0, 1, 0, 0, 0, 0],
                       [0, 0, 0, 0, -1, 0, 0, 0],
                       [0, 0, 0, 0, 0, 1, 0, 0],
                       [0, 0, 0, 0, 0, 0, 1, 0],
                       [0, 0, 0, 0, 0, 0, 0, 1],])
Operator.is_unitary(phase_oracle)
```

Out[4]: True

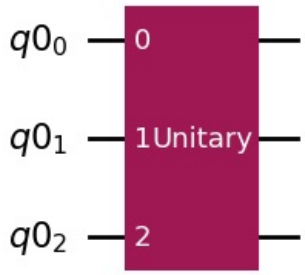
Task 2 : Gate

XOR Oracle

In [5]:
```
n0=3
q0 = QuantumRegister(n0)
XOR_circuit = QuantumCircuit(q0, name='Uf')
XOR_circuit.append(xor_oracle,[0, 1, 2])

XOR_gate = XOR_circuit.to_gate()
```

```
XOR_circuit.draw(output='mpl')
```

Out[5]:



In [6]:
```
XOR_gate
```

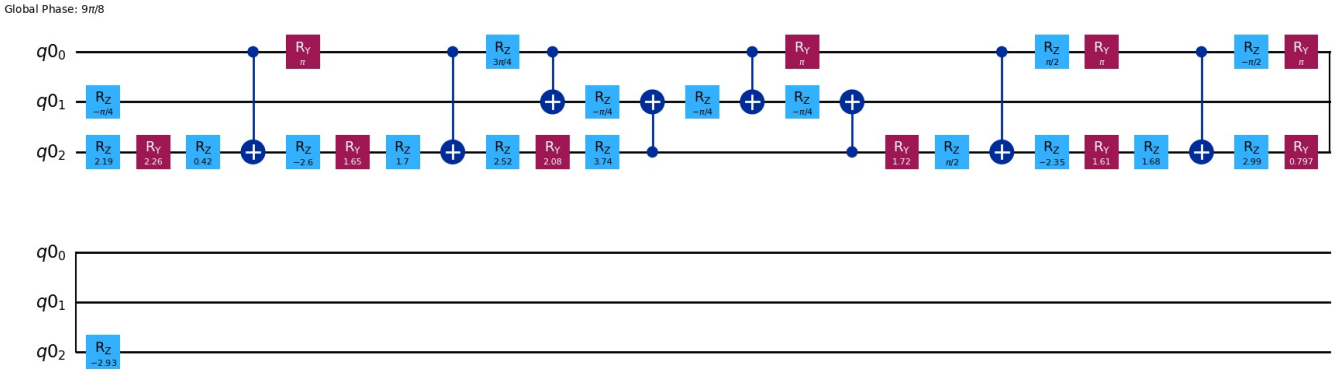
Out[6]:
```
 Instruction(name='Uf', num_qubits=3, num_clbits=0, params=[])
```

In [7]:
```
Operator(XOR_circuit)
```

```
Operator([[1.+0.j, 0.+0.j, 0.+0.j, 0.+0.j, 0.+0.j, 0.+0.j, 0.+0.j, 0.+0.j],
          [0.+0.j, 0.+0.j, 0.+0.j, 0.+0.j, 0.+0.j, 1.+0.j, 0.+0.j, 0.+0.j],
          [0.+0.j, 0.+0.j, 1.+0.j, 0.+0.j, 0.+0.j, 0.+0.j, 0.+0.j, 0.+0.j],
          [0.+0.j, 0.+0.j, 0.+0.j, 1.+0.j, 0.+0.j, 0.+0.j, 0.+0.j, 0.+0.j],
          [0.+0.j, 0.+0.j, 0.+0.j, 0.+0.j, 1.+0.j, 0.+0.j, 0.+0.j, 0.+0.j],
          [0.+0.j, 1.+0.j, 0.+0.j, 0.+0.j, 0.+0.j, 0.+0.j, 0.+0.j, 0.+0.j],
          [0.+0.j, 0.+0.j, 0.+0.j, 0.+0.j, 0.+0.j, 0.+0.j, 1.+0.j, 0.+0.j],
          [0.+0.j, 0.+0.j, 0.+0.j, 0.+0.j, 0.+0.j, 0.+0.j, 0.+0.j, 1.+0.j]],
         input_dims=(2, 2, 2), output_dims=(2, 2, 2))
```

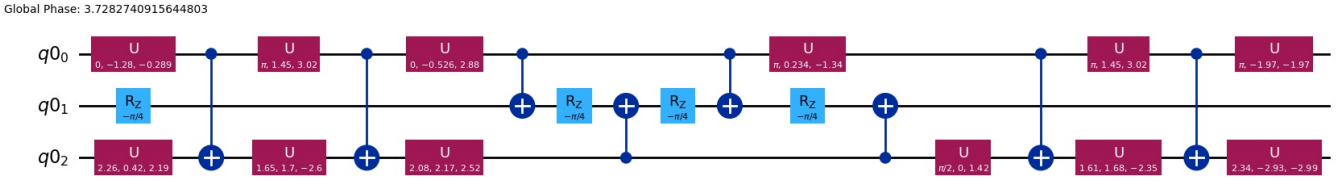
In [8]:
```
XOR_circuit_transpiled = transpile(XOR_circuit,  basis_gates=['cx', 'rz', 'ry', 'h'])
XOR_circuit_transpiled.draw(output='mpl')
```

Out[8]:



In [9]:
```
XOR_circuit_decomp=XOR_circuit.decompose(reps=2)
XOR_circuit_decomp.draw(output='mpl')
```

Out[9]:



In [10]:
```
circuit_to_run = XOR_circuit_transpiled.copy()
circuit_to_run.save_unitary()
job = backend.run(circuit_to_run)
results = job.result()
print(results.get_unitary(circuit_to_run, decimals=3))
```

```
Operator([[ 1.-0.j,  0.-0.j,  0.+0.j,  0.+0.j, -0.-0.j,  0.-0.j,  0.+0.j,
            0.+0.j],
          [-0.-0.j,  0.-0.j,  0.+0.j,  0.+0.j,  0.-0.j,  1.-0.j,  0.+0.j,
            0.+0.j],
          [ 0.+0.j,  0.+0.j,  1.-0.j,  0.-0.j,  0.+0.j,  0.+0.j,  0.-0.j,
            0.-0.j],
          [ 0.+0.j,  0.+0.j, -0.-0.j,  1.-0.j,  0.+0.j,  0.+0.j,  0.-0.j,
           -0.-0.j],
          [ 0.-0.j,  0.+0.j,  0.+0.j,  0.+0.j,  1.-0.j, -0.-0.j,  0.+0.j,
            0.+0.j],
          [-0.-0.j,  1.-0.j,  0.+0.j,  0.+0.j, -0.+0.j, -0.-0.j,  0.+0.j,
            0.+0.j],
          [ 0.+0.j,  0.+0.j,  0.-0.j, -0.+0.j,  0.+0.j,  0.+0.j,  1.-0.j,
            0.+0.j],
          [ 0.+0.j,  0.+0.j, -0.-0.j,  0.-0.j,  0.+0.j,  0.+0.j, -0.+0.j,
            1.-0.j]],
         input_dims=(2, 2, 2), output_dims=(2, 2, 2))
```

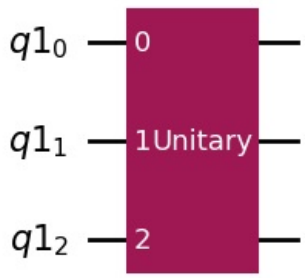
In [11]:
```
circuit_to_run = XOR_circuit_decomp.copy()
circuit_to_run.save_unitary()
job = backend.run(circuit_to_run)
results = job.result()
print(results.get_unitary(circuit_to_run, decimals=3))
```

```
Operator([[ 1.+0.j, -0.+0.j,  0.+0.j,  0.+0.j, -0.+0.j, -0.+0.j,  0.+0.j,
            0.+0.j],
          [ 0.+0.j,  0.+0.j,  0.+0.j,  0.+0.j, -0.-0.j,  1.+0.j,  0.+0.j,
            0.+0.j],
          [ 0.+0.j,  0.+0.j,  1.+0.j, -0.+0.j,  0.+0.j,  0.+0.j,  0.+0.j,
           -0.-0.j],
          [ 0.+0.j,  0.+0.j,  0.+0.j,  1.+0.j,  0.+0.j,  0.+0.j, -0.-0.j,
           -0.+0.j],
          [-0.+0.j, -0.+0.j,  0.+0.j,  0.+0.j,  1.+0.j,  0.-0.j,  0.+0.j,
            0.+0.j],
          [ 0.+0.j,  1.+0.j,  0.+0.j,  0.+0.j,  0.+0.j, -0.+0.j,  0.+0.j,
            0.+0.j],
          [ 0.+0.j,  0.+0.j,  0.+0.j,  0.-0.j,  0.+0.j,  0.+0.j,  1.+0.j,
           -0.+0.j],
          [ 0.+0.j,  0.+0.j,  0.-0.j,  0.+0.j,  0.+0.j,  0.+0.j,  0.+0.j,
            1.+0.j]],
         input_dims=(2, 2, 2), output_dims=(2, 2, 2))
```

Phase Oracle

In [12]:
```
n0=3
q0 = QuantumRegister(n0)
phase_circuit = QuantumCircuit(q0, name='Uf')
phase_circuit.append(phase_oracle,[0, 1, 2])

phase_gate = phase_circuit.to_gate()
phase_circuit.draw(output='mpl')
```

Out[12]:



In [13]:
```
phase_gate
```

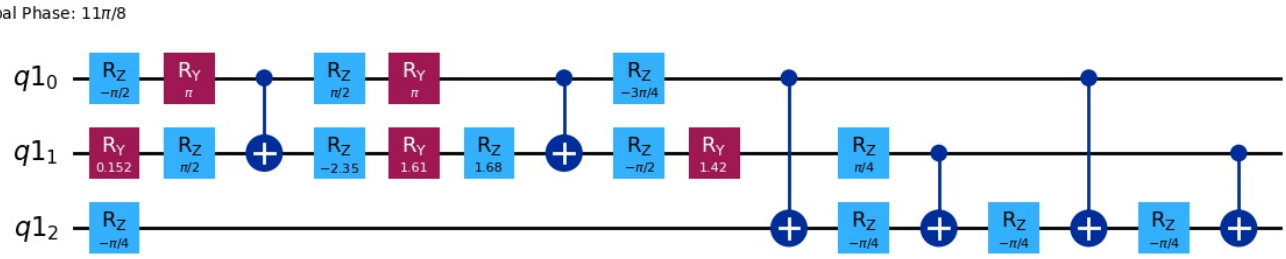
Out[13]:
```
Instruction(name='Uf', num_qubits=3, num_clbits=0, params=[])
```

In [14]:
```
Operator(phase_circuit)
```

```
Operator([[ 1.+0.j,  0.+0.j,  0.+0.j,  0.+0.j,  0.+0.j,  0.+0.j,  0.+0.j,
            0.+0.j],
          [ 0.+0.j,  1.+0.j,  0.+0.j,  0.+0.j,  0.+0.j,  0.+0.j,  0.+0.j,
            0.+0.j],
          [ 0.+0.j,  0.+0.j,  1.+0.j,  0.+0.j,  0.+0.j,  0.+0.j,  0.+0.j,
            0.+0.j],
          [ 0.+0.j,  0.+0.j,  0.+0.j,  1.+0.j,  0.+0.j,  0.+0.j,  0.+0.j,
            0.+0.j],
          [ 0.+0.j,  0.+0.j,  0.+0.j,  0.+0.j, -1.+0.j,  0.+0.j,  0.+0.j,
            0.+0.j],
          [ 0.+0.j,  0.+0.j,  0.+0.j,  0.+0.j,  0.+0.j,  1.+0.j,  0.+0.j,
            0.+0.j],
          [ 0.+0.j,  0.+0.j,  0.+0.j,  0.+0.j,  0.+0.j,  0.+0.j,  1.+0.j,
            0.+0.j],
          [ 0.+0.j,  0.+0.j,  0.+0.j,  0.+0.j,  0.+0.j,  0.+0.j,  0.+0.j,
            1.+0.j]],
         input_dims=(2, 2, 2), output_dims=(2, 2, 2))
```

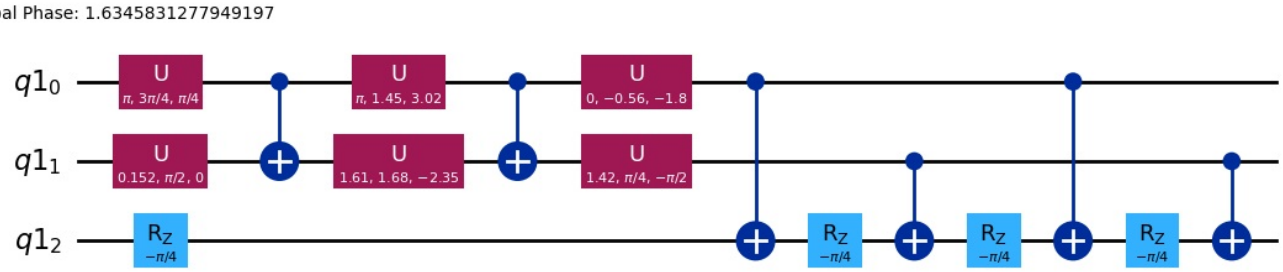
In [15]:
```
phase_circuit_transpiled = transpile(phase_circuit, basis_gates=['cx', 'rz', 'ry', 'h'])
phase_circuit_transpiled.draw(output='mpl')
```

Out[15]:



In [16]:
```
phase_circuit_decomp=phase_circuit.decompose(reps=2)
phase_circuit_decomp.draw(output='mpl')
```

Out[16]:



In [17]:
```
circuit_to_run = phase_circuit_transpiled.copy()
circuit_to_run.save_unitary()
job = backend.run(circuit_to_run)
results = job.result()
print(results.get_unitary(circuit_to_run, decimals=3))
```

```
Operator([[ 1.-0.j, -0.-0.j,  0.-0.j, -0.-0.j,  0.+0.j,  0.+0.j,  0.+0.j,
            0.+0.j],
          [-0.-0.j,  1.-0.j,  0.+0.j, -0.-0.j,  0.+0.j,  0.+0.j,  0.+0.j,
            0.+0.j],
          [-0.-0.j,  0.+0.j,  1.-0.j, -0.-0.j,  0.+0.j,  0.+0.j,  0.+0.j,
            0.+0.j],
          [-0.-0.j,  0.-0.j, -0.-0.j,  1.-0.j,  0.+0.j,  0.+0.j,  0.+0.j,
            0.+0.j],
          [ 0.+0.j,  0.+0.j,  0.+0.j,  0.+0.j, -1.+0.j,  0.+0.j, -0.+0.j,
            0.+0.j],
          [ 0.+0.j,  0.+0.j,  0.+0.j,  0.+0.j, -0.-0.j,  1.-0.j,  0.+0.j,
           -0.-0.j],
          [ 0.+0.j,  0.+0.j,  0.+0.j,  0.+0.j, -0.-0.j,  0.+0.j,  1.-0.j,
           -0.-0.j],
          [ 0.+0.j,  0.+0.j,  0.+0.j,  0.+0.j, -0.-0.j,  0.-0.j, -0.-0.j,
            1.-0.j]],
         input_dims=(2, 2, 2), output_dims=(2, 2, 2))
```

In [18]:
```
circuit_to_run = phase_circuit_decomp.copy()
circuit_to_run.save_unitary()
job = backend.run(circuit_to_run)
results = job.result()
print(results.get_unitary(circuit_to_run, decimals=3))
```

```
Operator([[ 1.+0.j,  0.-0.j,  0.-0.j, -0.-0.j,  0.+0.j,  0.+0.j,  0.+0.j,
            0.+0.j],
          [-0.-0.j,  1.+0.j, -0.+0.j,  0.-0.j,  0.+0.j,  0.+0.j,  0.+0.j,
            0.+0.j],
          [-0.-0.j,  0.+0.j,  1.+0.j, -0.-0.j,  0.+0.j,  0.+0.j,  0.+0.j,
            0.+0.j],
          [ 0.-0.j, -0.-0.j,  0.-0.j,  1.+0.j,  0.+0.j,  0.+0.j,  0.+0.j,
            0.+0.j],
          [ 0.+0.j,  0.+0.j,  0.+0.j,  0.+0.j, -1.+0.j, -0.+0.j, -0.+0.j,
            0.+0.j],
          [ 0.+0.j,  0.+0.j,  0.+0.j,  0.+0.j, -0.-0.j,  1.+0.j, -0.+0.j,
            0.-0.j],
          [ 0.+0.j,  0.+0.j,  0.+0.j,  0.+0.j, -0.-0.j,  0.+0.j,  1.+0.j,
           -0.-0.j],
          [ 0.+0.j,  0.+0.j,  0.+0.j,  0.+0.j,  0.-0.j, -0.-0.j,  0.-0.j,
            1.+0.j]],
         input_dims=(2, 2, 2), output_dims=(2, 2, 2))
```

Task 3 :

In [19]:
```python
n = 3
dim = 2**n

ket0 = np.zeros((dim, 1))
ket0[0, 0] = 1

proj = ket0 @ ket0.T

print(proj)
```

```
[[1. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0.]]
```

Task 4 :

In [20]:
```python
n = 3
dim = 2**n

ket0 = np.zeros((dim, 1))
ket0[0, 0] = 1
proj = ket0 @ ket0.T
sol = 2 * ket0 @ ket0.T - np.eye(proj.shape[0], proj.shape[1])
print(sol)
```

```
[[ 1.  0.  0.  0.  0.  0.  0.  0.]
 [ 0. -1.  0.  0.  0.  0.  0.  0.]
 [ 0.  0. -1.  0.  0.  0.  0.  0.]
 [ 0.  0.  0. -1.  0.  0.  0.  0.]
 [ 0.  0.  0.  0. -1.  0.  0.  0.]
 [ 0.  0.  0.  0.  0. -1.  0.  0.]
 [ 0.  0.  0.  0.  0.  0. -1.  0.]
 [ 0.  0.  0.  0.  0.  0.  0. -1.]]
```

Task 5 :

In [21]:
```python
n = 3
dim = 2**n

ket0 = np.zeros((dim, 1))
ket0[0, 0] = 1
proj = ket0 @ ket0.T
X_1 = np.array([[0, 1],
                [1, 0]])
X_n = X_1
for i in range(1, n):
    X_n = np.kron(X_n, X_1)

lhs = X_n @ (2 * ket0 @ ket0.T - np.eye(proj.shape[0], proj.shape[1])) @ X_n

ket1 = np.zeros((dim, 1))
ket1[-1, -1] = 1
rhs = 2 * ket1 @ ket1.T - np.eye(proj.shape[0], proj.shape[1])
print(f"Left Hand Side:\n {lhs}")
print(f"Right Hand Side:\n {rhs}")
```

```
print(f"Equal: {np.allclose(lhs, rhs)}")
```

```
Left Hand Side:
 [[-1.  0.  0.  0.  0.  0.  0.  0.]
 [ 0. -1.  0.  0.  0.  0.  0.  0.]
 [ 0.  0. -1.  0.  0.  0.  0.  0.]
 [ 0.  0.  0. -1.  0.  0.  0.  0.]
 [ 0.  0.  0.  0. -1.  0.  0.  0.]
 [ 0.  0.  0.  0.  0. -1.  0.  0.]
 [ 0.  0.  0.  0.  0.  0. -1.  0.]
 [ 0.  0.  0.  0.  0.  0.  0.  1.]]
Right Hand Side:
 [[-1.  0.  0.  0.  0.  0.  0.  0.]
 [ 0. -1.  0.  0.  0.  0.  0.  0.]
 [ 0.  0. -1.  0.  0.  0.  0.  0.]
 [ 0.  0.  0. -1.  0.  0.  0.  0.]
 [ 0.  0.  0.  0. -1.  0.  0.  0.]
 [ 0.  0.  0.  0.  0. -1.  0.  0.]
 [ 0.  0.  0.  0.  0.  0. -1.  0.]
 [ 0.  0.  0.  0.  0.  0.  0.  1.]]
Equal: True
```

Task 6 : Grover's Diffusion Operator

In [22]:
```
n = 3
dim = 2**n

phi = 1/sqrt(dim) * np.ones((dim, 1))
W = 2 * phi @ phi.T - np.eye(dim, dim)
print(W)
```

```
[[-0.75  0.25  0.25  0.25  0.25  0.25  0.25  0.25]
 [ 0.25 -0.75  0.25  0.25  0.25  0.25  0.25  0.25]
 [ 0.25  0.25 -0.75  0.25  0.25  0.25  0.25  0.25]
 [ 0.25  0.25  0.25 -0.75  0.25  0.25  0.25  0.25]
 [ 0.25  0.25  0.25  0.25 -0.75  0.25  0.25  0.25]
 [ 0.25  0.25  0.25  0.25  0.25 -0.75  0.25  0.25]
 [ 0.25  0.25  0.25  0.25  0.25  0.25 -0.75  0.25]
 [ 0.25  0.25  0.25  0.25  0.25  0.25  0.25 -0.75]]
```

In [23]:
```
n = 3
dim = 2**n

ket0 = np.zeros((dim, 1))
ket0[0, 0] = 1
H_1 = 1/sqrt(2) * np.array([[1, 1],
                            [1, -1]])
H_n = H_1
for i in range(1, n):
    H_n = np.kron(H_n, H_1)

W = H_n @ (2* ket0 @ ket0.T - np.eye(dim, dim)) @ H_n
print(W)
```

```
[[-0.75  0.25  0.25  0.25  0.25  0.25  0.25  0.25]
 [ 0.25 -0.75  0.25  0.25  0.25  0.25  0.25  0.25]
 [ 0.25  0.25 -0.75  0.25  0.25  0.25  0.25  0.25]
 [ 0.25  0.25  0.25 -0.75  0.25  0.25  0.25  0.25]
 [ 0.25  0.25  0.25  0.25 -0.75  0.25  0.25  0.25]
 [ 0.25  0.25  0.25  0.25  0.25 -0.75  0.25  0.25]
 [ 0.25  0.25  0.25  0.25  0.25  0.25 -0.75  0.25]
 [ 0.25  0.25  0.25  0.25  0.25  0.25  0.25 -0.75]]
```

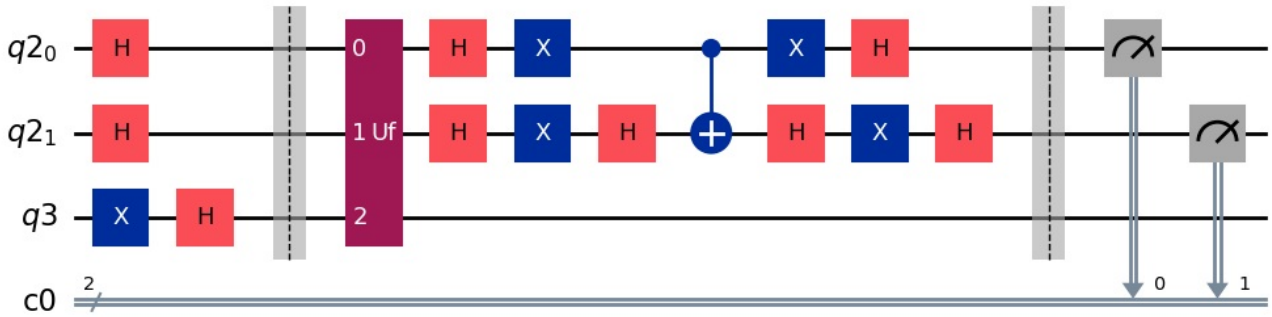
Task 7 :

XOR Operator

In [24]:
```
n0 = 3
q0 = QuantumRegister(n0 - 1)
ancilla = QuantumRegister(1)
c0 = ClassicalRegister(n0 - 1)
W_XOR_circuit = QuantumCircuit(q0, ancilla, c0, name='W_XOR')

W_XOR_circuit.x(ancilla)
W_XOR_circuit.h(ancilla)
W_XOR_circuit.h(q0)
W_XOR_circuit.barrier()

num_iter = int(pi/4 * sqrt(2**(n0-1)))

for i in range(num_iter):
```

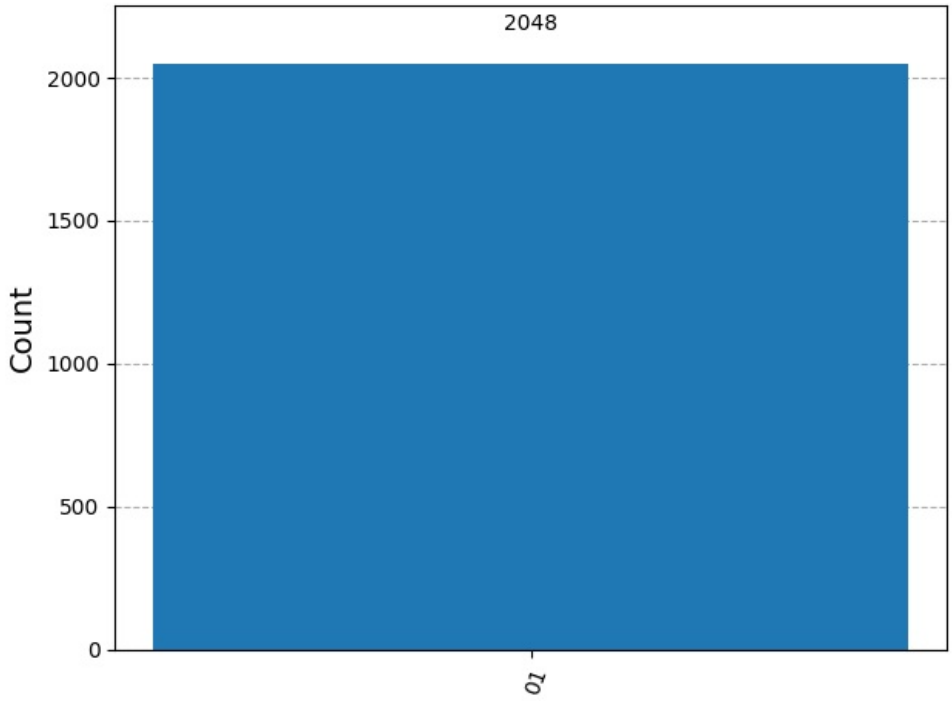
```
    W_XOR_circuit.append(XOR_gate, [q0[0], q0[1], ancilla])
    W_XOR_circuit.h(q0)
    W_XOR_circuit.x(q0)
    W_XOR_circuit.h(q0[1])
    W_XOR_circuit.cx(q0[0], q0[1])
    W_XOR_circuit.h(q0[1])
    W_XOR_circuit.x(q0)
    W_XOR_circuit.h(q0)
    W_XOR_circuit.barrier()

W_XOR_circuit.measure(q0, c0)
W_XOR_circuit.draw(output='mpl')
```

Out[24]:



In [25]:
```
W_XOR_circuit_transpiled = transpile(W_XOR_circuit, experiment_backend)
job = experiment_backend.run(W_XOR_circuit_transpiled, shots=2048)
result = job.result()
counts = result.get_counts()
plot_histogram(counts)
```

Out[25]:

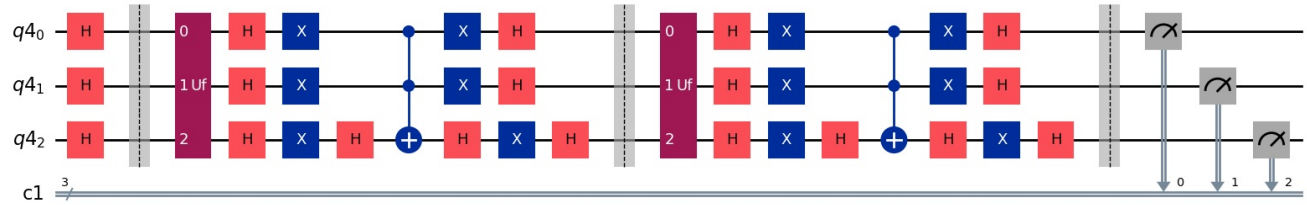


Phase Operator

In [26]:
```
n0 = 3
q0 = QuantumRegister(n0)
c0 = ClassicalRegister(n0)
W_Phase_circuit = QuantumCircuit(q0, c0, name='W_XOR')

W_Phase_circuit.h(q0)
W_Phase_circuit.barrier()

num_iter = int(pi/4 * sqrt(2**n0))

for i in range(num_iter):
    W_Phase_circuit.append(phase_gate, [0, 1, 2])
    W_Phase_circuit.h(q0)
    W_Phase_circuit.x(q0)
    W_Phase_circuit.h(q0[2])
    W_Phase_circuit.ccx(q0[0], q0[1], q0[2])
    W_Phase_circuit.h(q0[2])
```

```
    W_Phase_circuit.x(q0)
    W_Phase_circuit.h(q0)
    W_Phase_circuit.barrier()

W_Phase_circuit.measure(q0, c0)
W_Phase_circuit.draw(output='mpl')
```

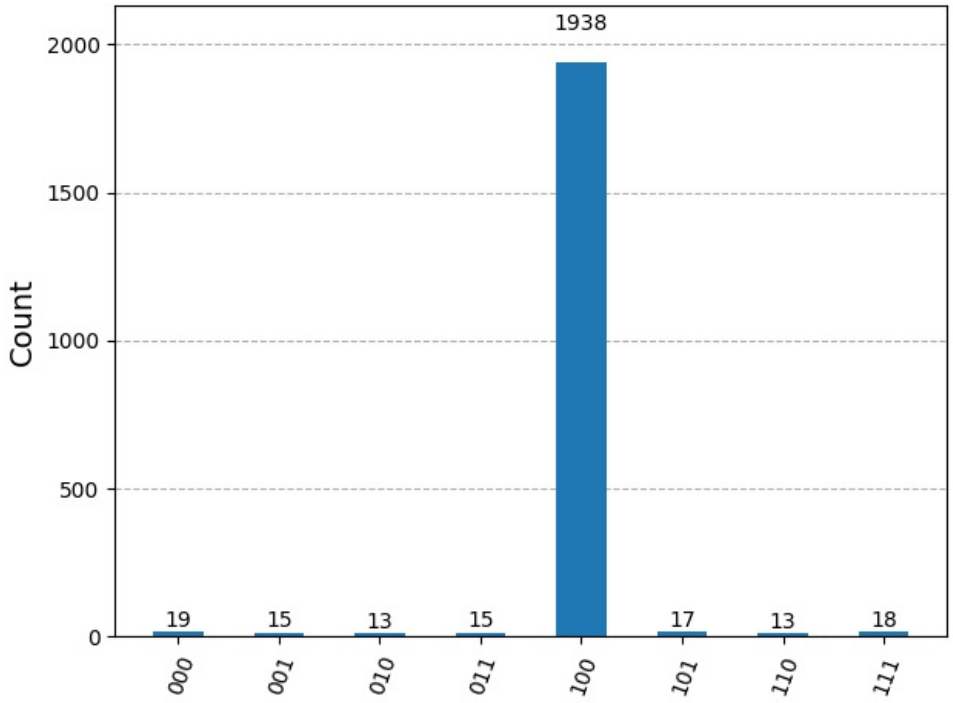
Out[26]:



In [27]:

```
W_Phase_circuit_transpiled = transpile(W_Phase_circuit, experiment_backend)
job = experiment_backend.run(W_Phase_circuit_transpiled, shots=2048)
result = job.result()
counts = result.get_counts()
plot_histogram(counts)
```

Out[27]:

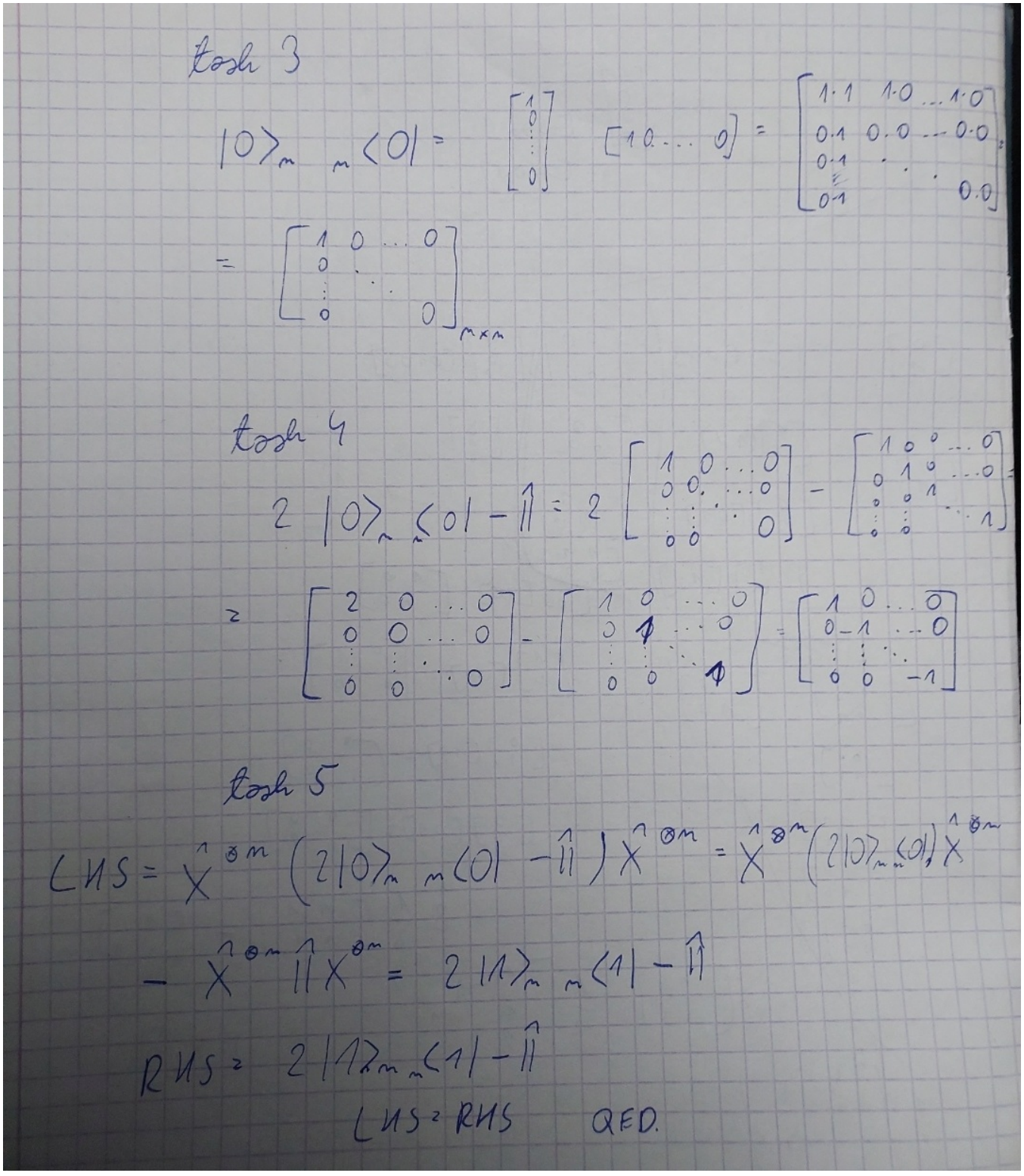


Tasks 3-6 (Manual Calculations)

In [1]:

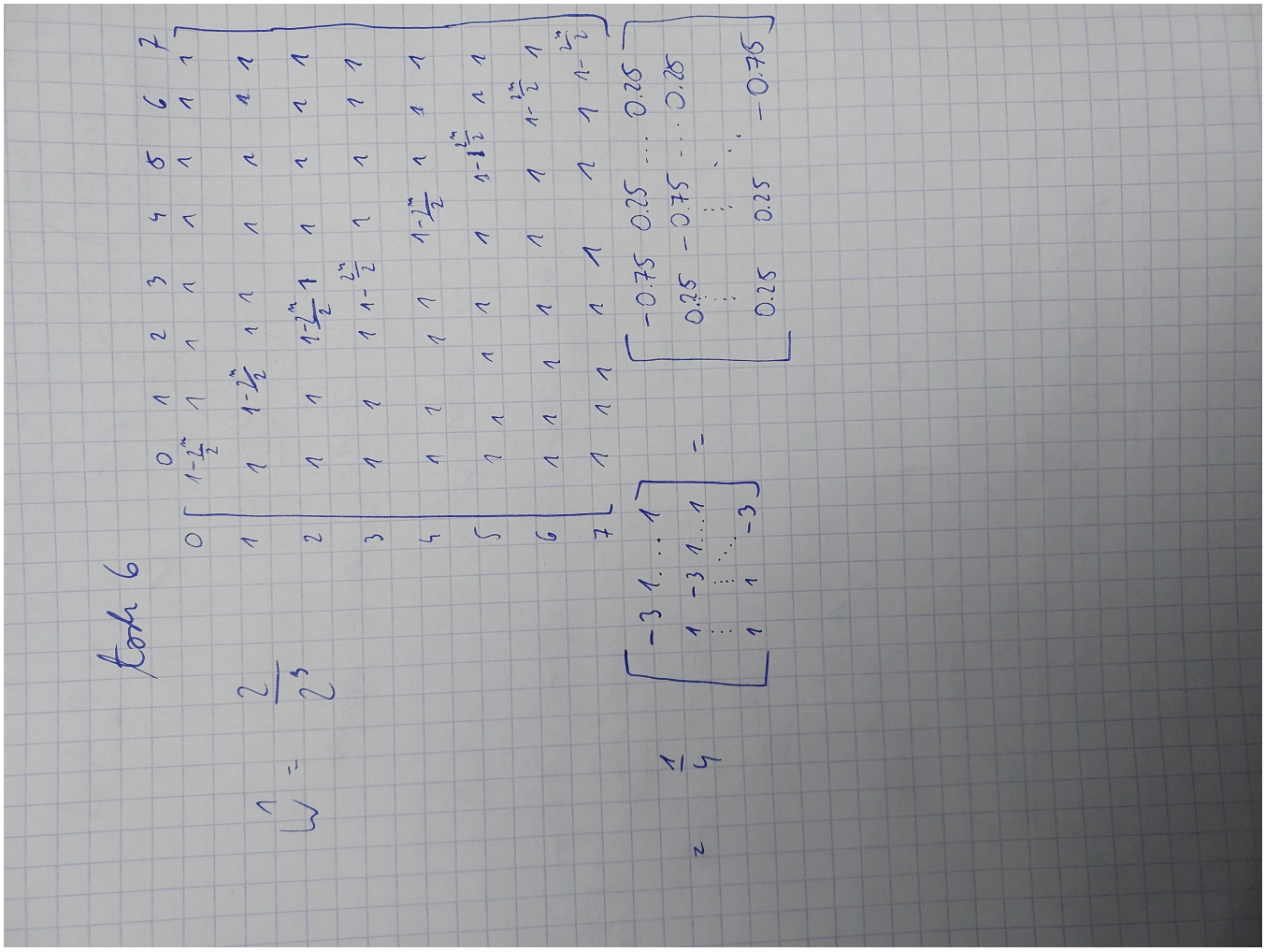
```
from IPython.display import Image
Image(filename="images/tasks3-5.jpg")
```

Out[1]:



In [2]:
```
Image(filename="images/task6.jpg")
```

Out[2]:



Conclusions

The project showcased a successfully construction and implementation of the XOR oracle, the phase oracle, and the Grover diffusion operator for a 3-qubit system. The circuits produced the expected unitaries and matched the theoretical formulas. The exercise clearly demonstrates how Grover’s algorithm can be assembled directly from its mathematical definitions.